

## Kafka Execution ( Kafka version – 4.1.1)

(for new creation start from step 1 else proceed from step-3)

### 1. Create new Cluster ID

1.Open Command Prompt (cmd) and navigate to your Kafka directory:

```
cd C:\kafka
```

2.Run the following command to generate a unique cluster UUID

```
.\bin\windows\kafka-storage.sh random-uuid
```

3.Copy the generated UUID string that appears in your console

### 2. Format Log Directories

1.Use the copied UUID to format the **directories (replace <your-cluster-id> with the UUID you copied)**:

```
bin\windows\kafka-storage.sh format --standalone -t <your-cluster-id> -c config\kraft\server.properties
```

2.Copy the generated UUID from the output and use it to replace <your-uuid-here> in the following command

```
.\bin\windows\kafka-storage.sh format --standalone -t <your-uuid-here> -c .\config\server.properties
```

### 3. Start the Kafka Server (applicable for both new/ existing kafka server)

1. Ensure the kafka post is uncommented the **server.properties** file in config & under windows folder of bin

```
listeners=PLAINTEXT://:9092
```

**Note:** edit the port number in above file if you need to change the port number

2.Check the below format also in same server.properties file and modify the port number which you have given in above place only for listener.

```
listeners=PLAINTEXT://localhost:9092,CONTROLLER://localhost:9093
```

Note:

**PLAINTEXT://localhost:9092 (Client Traffic)**

- **Purpose:** This is the port where external applications (like your **Spring Boot** apps, producers, or consumers) connect to send and receive data.
- **Protocol:** Unencrypted data (PLAINTEXT).

**CONTROLLER://localhost:9093 (Cluster Traffic)**

- **Purpose:** This is the port used internally by Kafka nodes to communicate with each other to manage cluster metadata, replacing the old ZooKeeper communication.
- **Protocol:** Internal metadata synchronization.

2.Run the below command to start the kafka server with port number mentioned in above step for starting kafka server

**bin\windows\kafka-server-start.bat config\server.properties**

**4.Create new topic: (Not required everytime, If you want to use existing topic skip this step)**

Run below command to create a new topic

**bin\windows\kafka-topics.bat --create --topic <my-first-topic> --bootstrap-server localhost:<portnumber where server is started> --partitions 1 --replication-factor 1**

**5.Check created or existing Topics:**

**bin\windows\Kafka-topics.bat --describe --topic <topic-name> --bootstrap-server localhost:<portnumber where server is started>**

**6.List all topics created:**

**bin\windows\Kafka-topics.bat --list --bootstrap-server localhost:<portnumber where server is started>**

Now you can run the application once after ensuring that the server is start and running